

3. Read the information about seagrass and use it to answer the questions.



Seagrass growing on the sea bed

Seagrass is a green plant which lives in the sea. It covers large areas of the sea bed.

Human activities such as dredging, building and industrial development can destroy seagrass. In the 1950s the loss of seagrass was 1.5 % per year but since 1990 scientists have calculated the loss to be 7 % per year.

Fisheries need sea grass. In the Mediterranean Sea, between 30 % and 40 % of the fish caught feed on seagrass and in Indonesian seas this is 60 %.

Seagrass is important in the carbon cycle, taking in carbon dioxide to produce glucose. This is converted into another carbon compound which can be stored in its large roots. Carbon dioxide makes sea water acidic which is harmful to sea organisms.

- (a) Answer the following questions using only the information given.

- (i) Give **one** reason why areas of seagrass are lost. [1]

- (ii) Which of the following statements describes the rate at which seagrass has been lost since 1950? [1]

- A** It has been constant every year
B It has increased
C It has decreased
D It has increased in some years and decreased in others.

Answer

(b) Answer the following questions using only the information given.

(i) Which statement describes how much the fishing industry depends on seagrass? [1]

- A Mediterranean and Indonesian fisheries depend totally on seagrass.
- B Mediterranean fisheries depend on seagrass for most of their fish.
- C Indonesian fisheries depend on seagrass for less than half of their fish.
- D Indonesian and Mediterranean fisheries depend on seagrass for some of their fish.

Answer

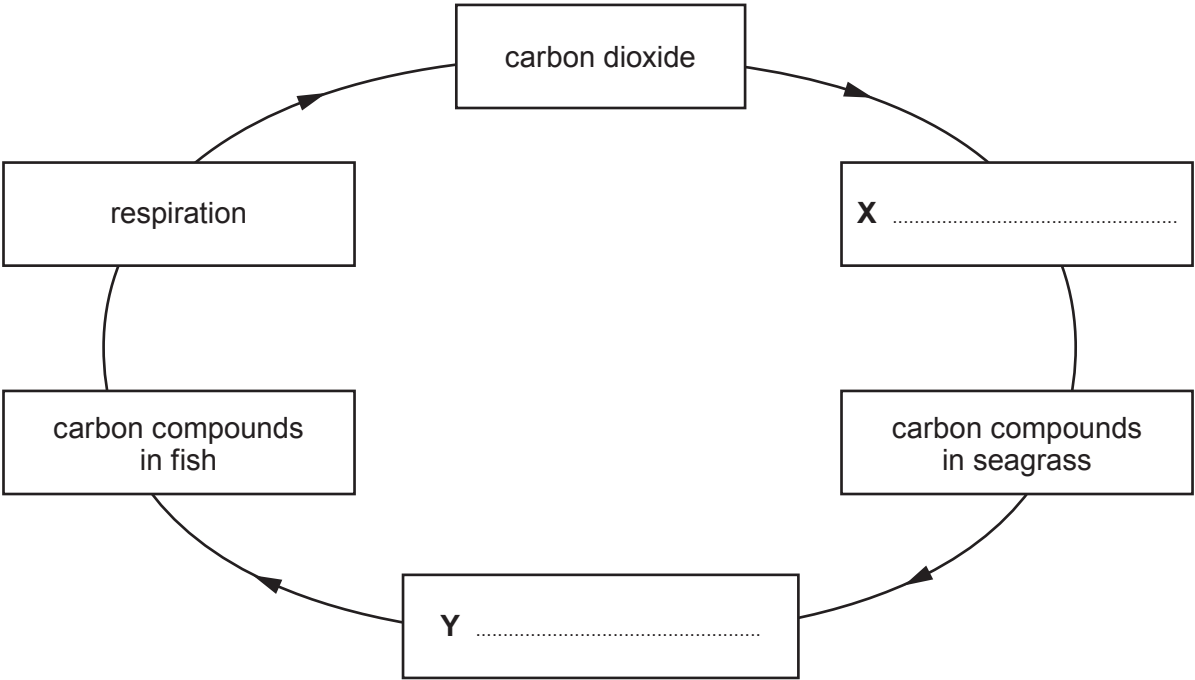
(ii) Apart from providing food, explain how seagrass is helpful to other species which live in the sea. [2]

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(c) Answer the following questions using the information given and your own knowledge.

(i) Complete the diagram below which shows part of the carbon cycle, by naming processes X and Y. [2]



(ii) Name the carbon compound stored in the roots of seagrass. [1]

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